

BTEC Level 3 Applied Science · Unit 1 Chemistry · Answer sheet

Atomic Structure — Answers

Mark scheme for the companion worksheet. Accept equivalent correct wording.

A

1. Proton +1, mass 1. Neutron 0, mass 1. Electron -1, mass $\approx 1/1836$ (very small).
2. Atomic number = number of protons. Mass number = protons + neutrons.
3. Same proton number / element; different neutron number / mass number.

B

4. 19 p, 20 n, 19 e.
5. Number of protons equals number of electrons, so charges cancel.
6. Mass is in the nucleus (protons and neutrons). Electrons occupy a much larger volume of mostly empty space.

C

7. Mg: 12p 12n 12e. Mg^{2+} : 12p 12n 10e. F: 9p 10n 9e. F^- : 9p 10n 10e.
8. $A_r = (60 \times 69 + 40 \times 71) / 100 = 69.8$
9. Chlorine is a mixture of ^{35}Cl and ^{37}Cl ; A_r is a weighted mean.

D

10. Isotopes must have the same proton number. C has 6 protons; N has 7, so they are different elements.
11. $(75 \times 63 + 25 \times 65) / 100 = 63.5$
12. The atom gains two electrons. Proton number is unchanged. Electrons now exceed protons by 2, so charge is 2-.

E

13. $A_r = (90.5 \times 20 + 0.27 \times 21 + 9.25 \times 22) / 100 = 20.2$ (3 s.f.). Ignoring ^{21}Ne changes the value only at the third significant figure / has a very small effect.